

STAT6 Appears to Be a Potential Regulator of Galectin-3 in Tauopathy

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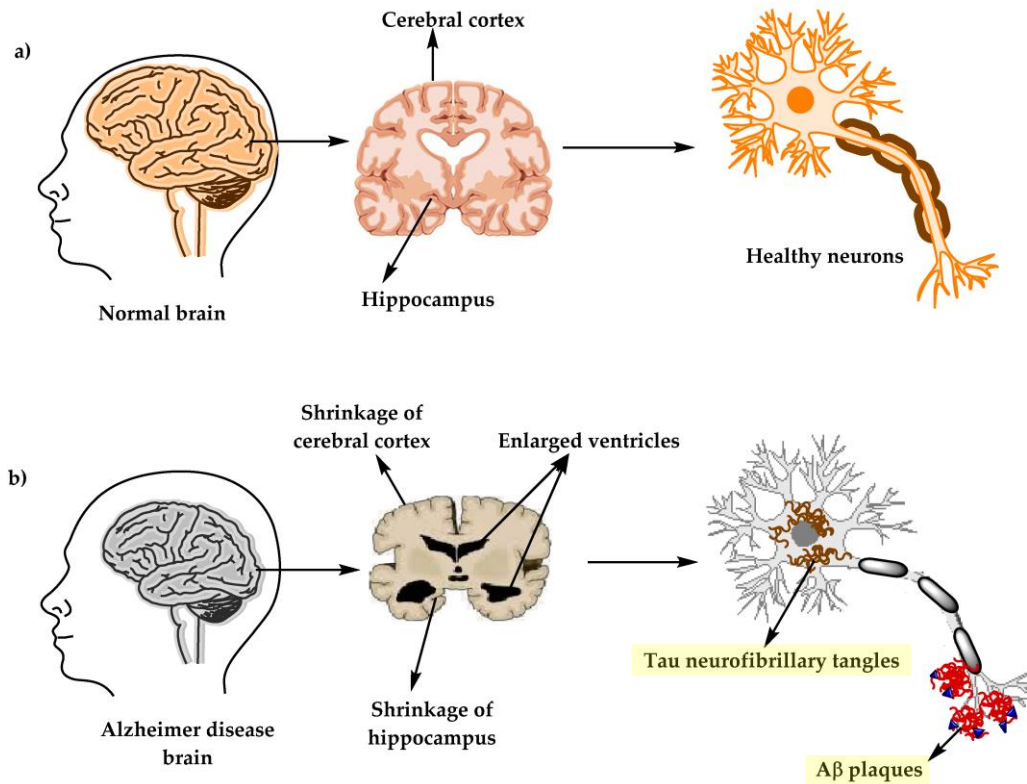
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Alzheimer's Disease

- Alzheimer's disease (AD) is a common neurodegenerative disease and a form of dementia, characterized by impaired cognition and behavior in elderly people over 65 years old.

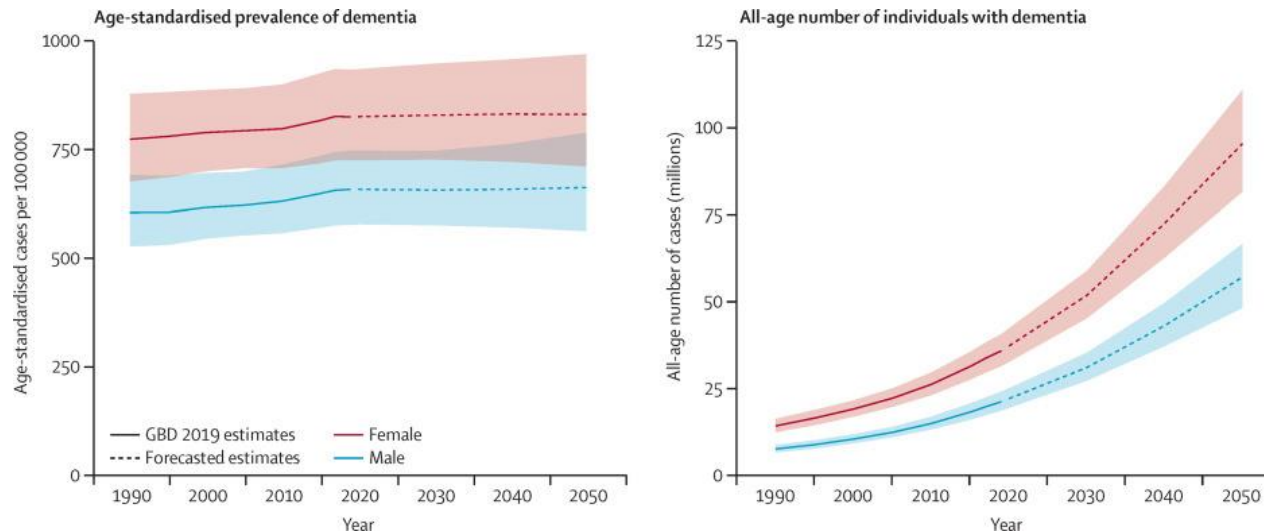
Alzheimer's Association. (2022). *Alzheimers Dement.*, 18.



Breijyeh & Karaman. (2020). *Molecules*, 25(24), 5789.

Alzheimer's Disease

- It is estimated that the number of people with dementia worldwide would increase from **57.4 million** cases in 2019 to 152.8 million cases in 2050.



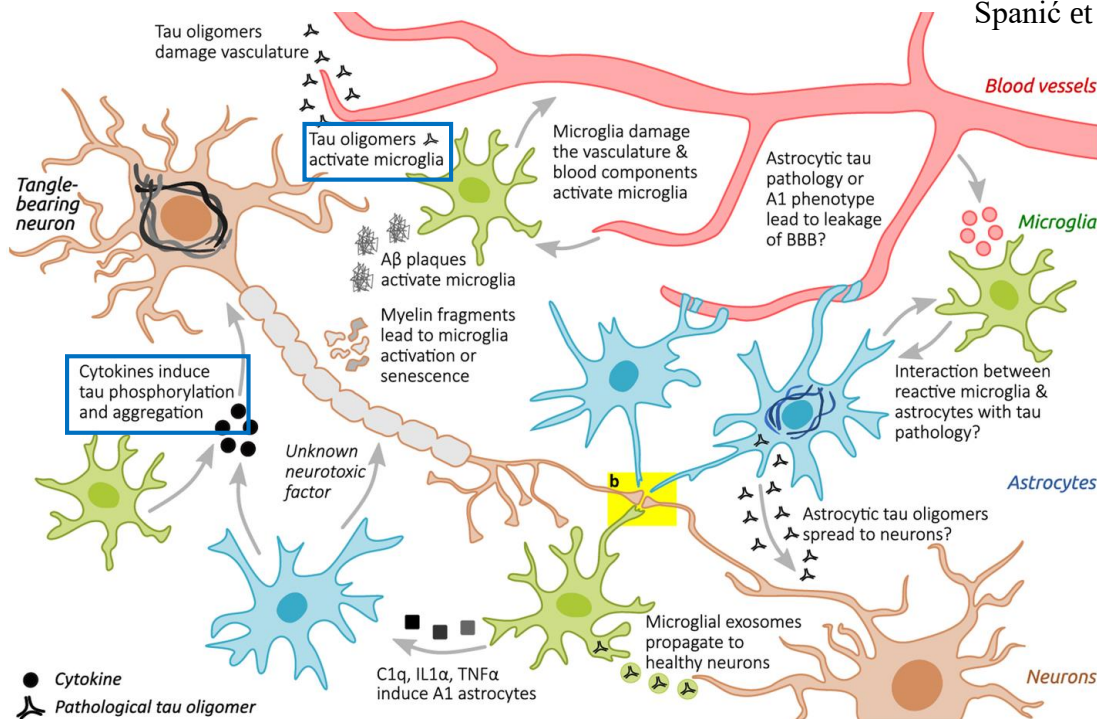
- In 2019, the estimated global cost of dementia was **US\$ 1.3 trillion**.

GBD 2019 Dementia Forecasting Collaborators (2022). *The Lancet. Public health*, 7(2), e105.
WHO. (2021). Dementia.

It is urgent to find solutions to Alzheimer's Disease.

Microglia

- Microglia cells are the resident macrophages of the CNS which can engulf over-generated neurons.
- In **AD**, **microglia** are continuously **activated**.
- **Tau** can **induce stronger microglial activation**, which in turn may promote the propagation of tau pathology.



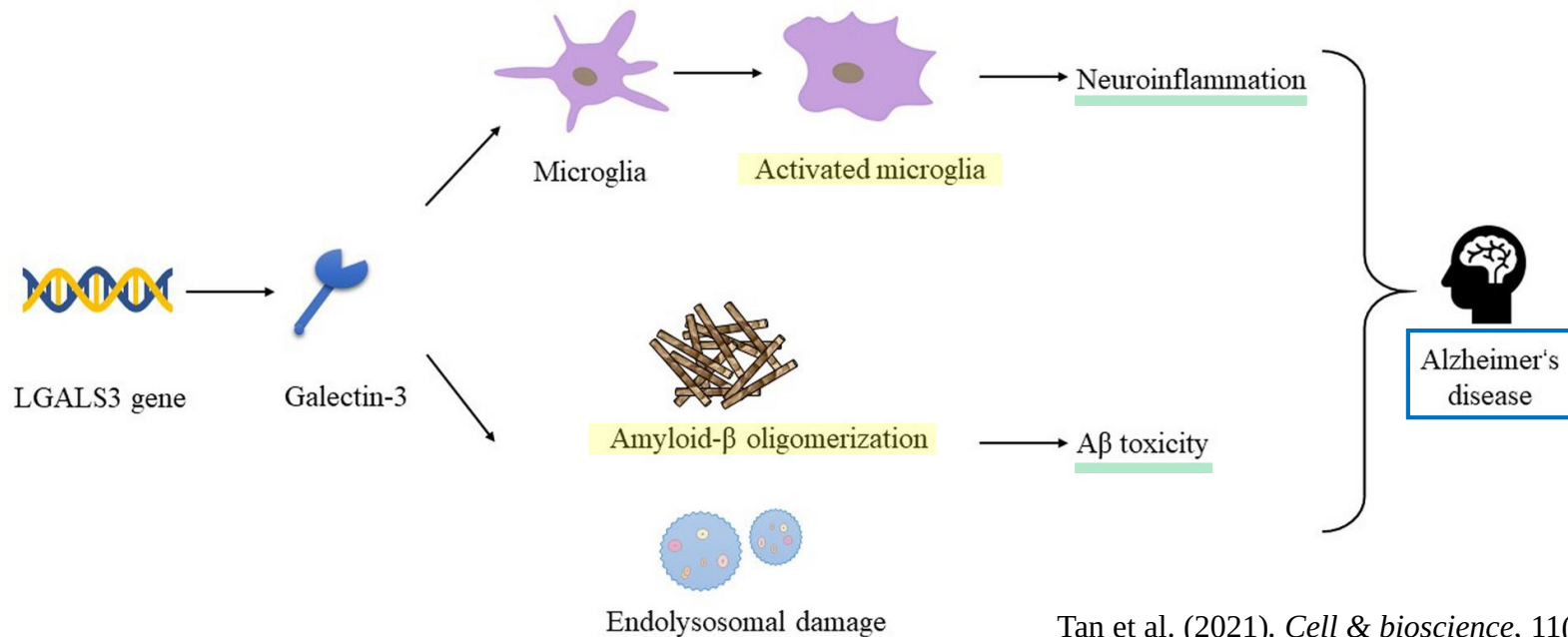
Perea et al. (2020). *Biomolecules*, 10(10), 1439.
 Španić et al. (2019). *Front. Aging Neurosci.*, 11, 271.

Vogels et al. (2019).
Acta Neuropathol. Commun., 7, 109.

Galectin-3 in Alzheimer's Disease

- **Galectin-3 (Gal3)** is highly **up-regulated** in brain cortex and glial cells **in AD** patients and AD mouse models.

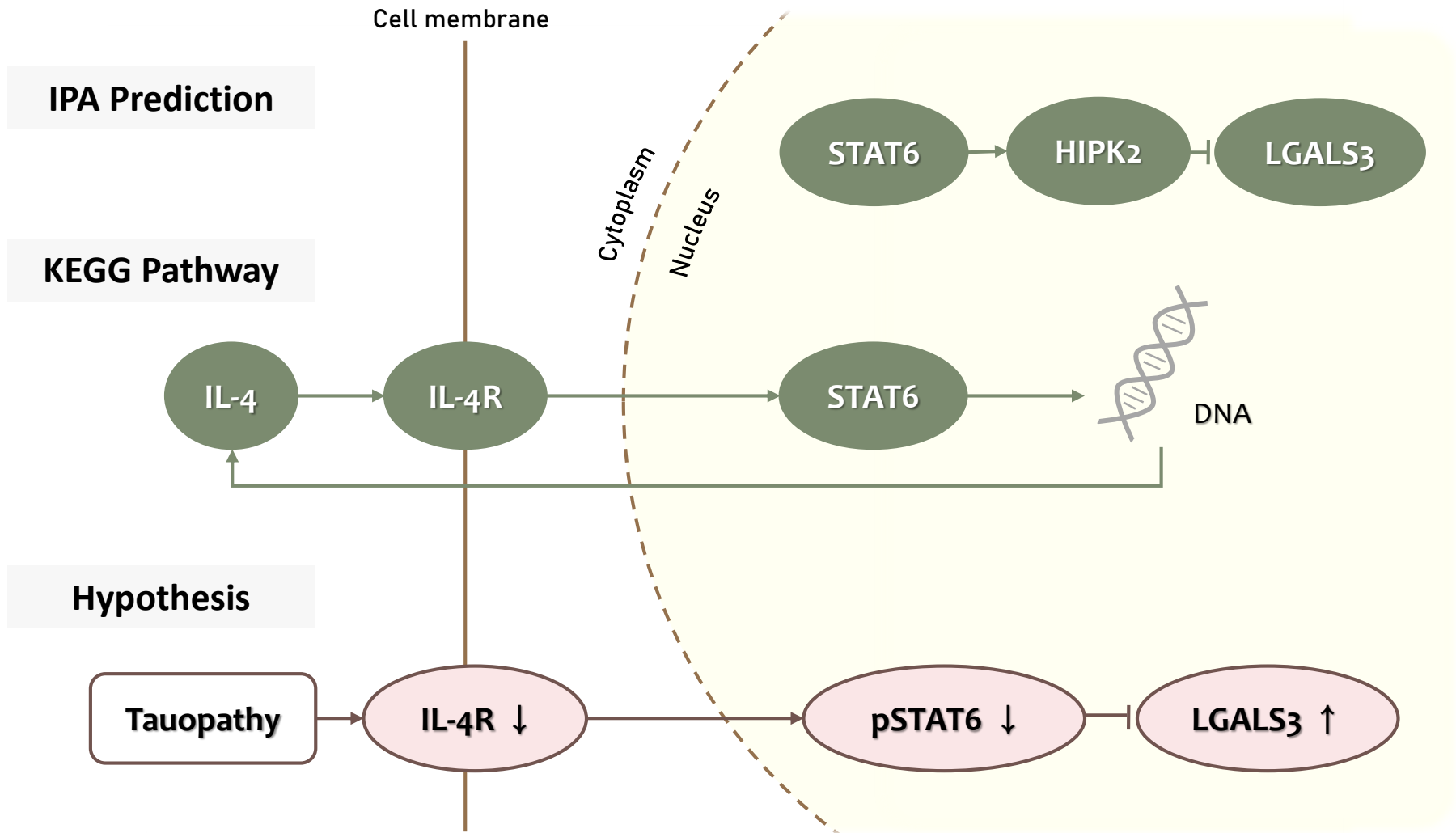
Puigdemívol et al. (2020). *Front. Cell Neurosci.*, 14, 162.



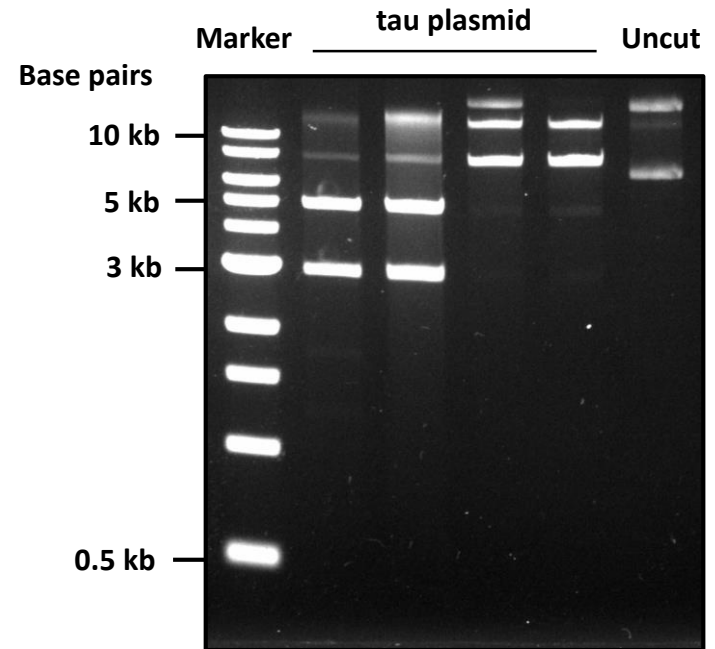
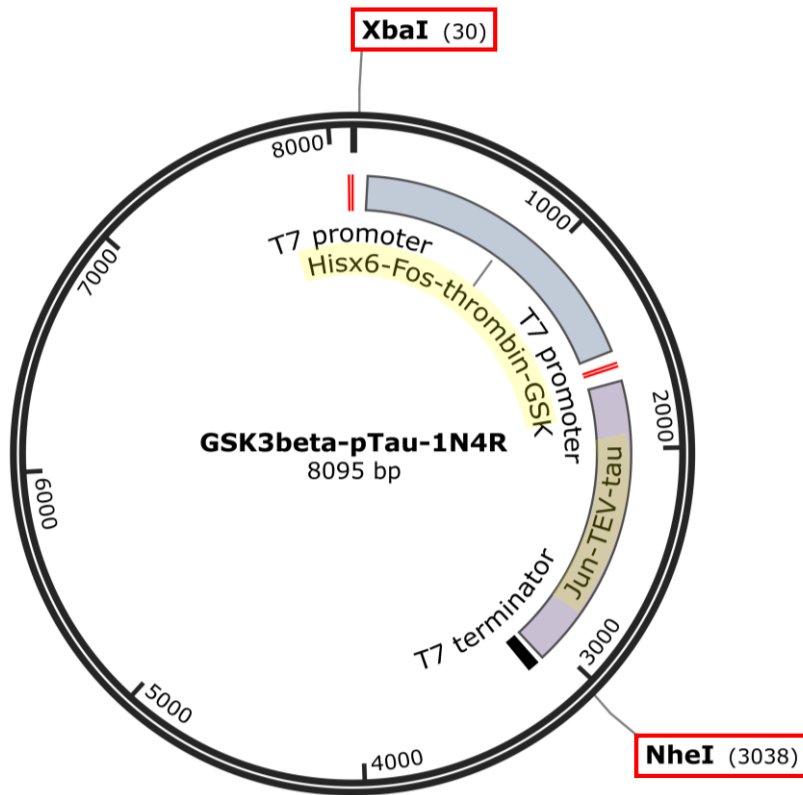
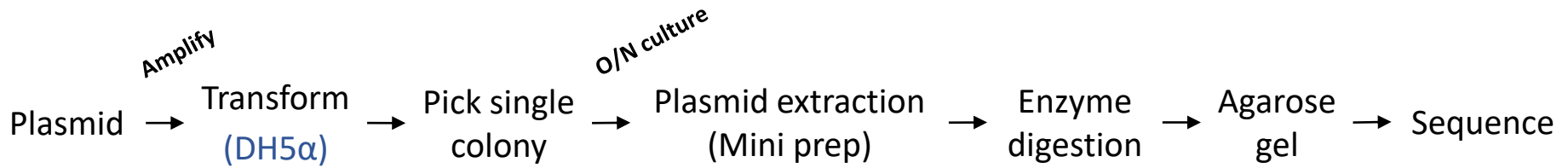
Tan et al. (2021). *Cell & bioscience*, 11(1), 78.

Galectin-3 is a potential therapeutic target in AD treatment.

Hypothesis of Galectin-3 Regulator in Tauopathy

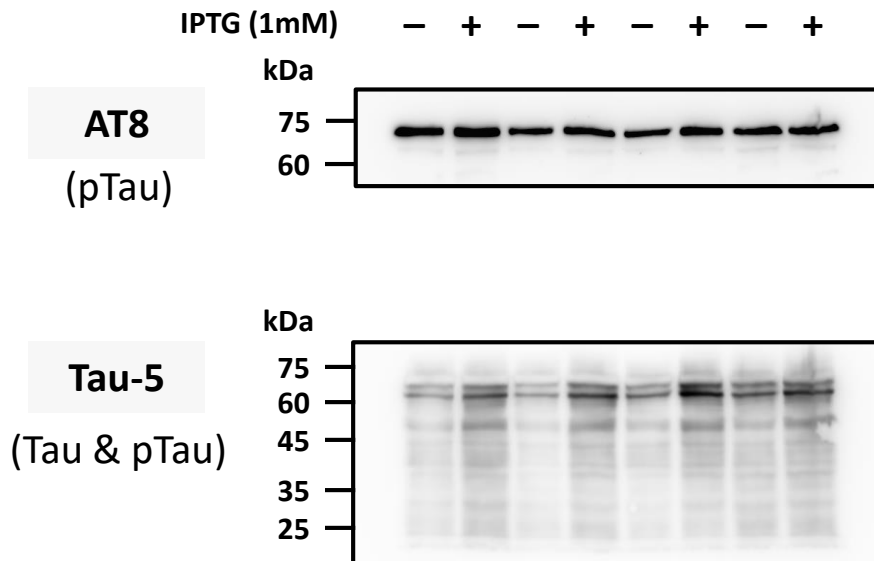
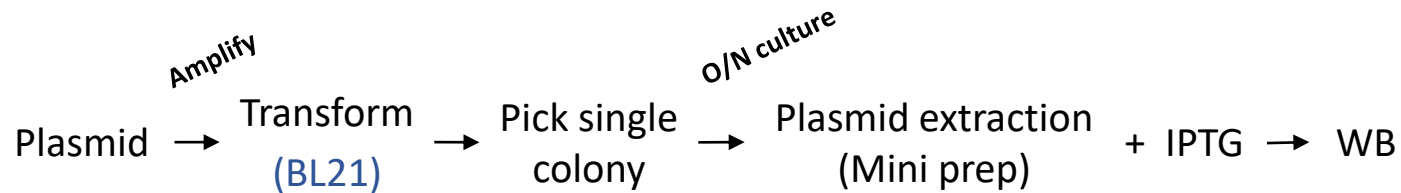


PIMAX Tau Plasmid



Tau protein
Predict size: 3k, 5kbp

Production of pTau and Tau Proteins: IPTG can induce BL21 PIMAX Tau Protein



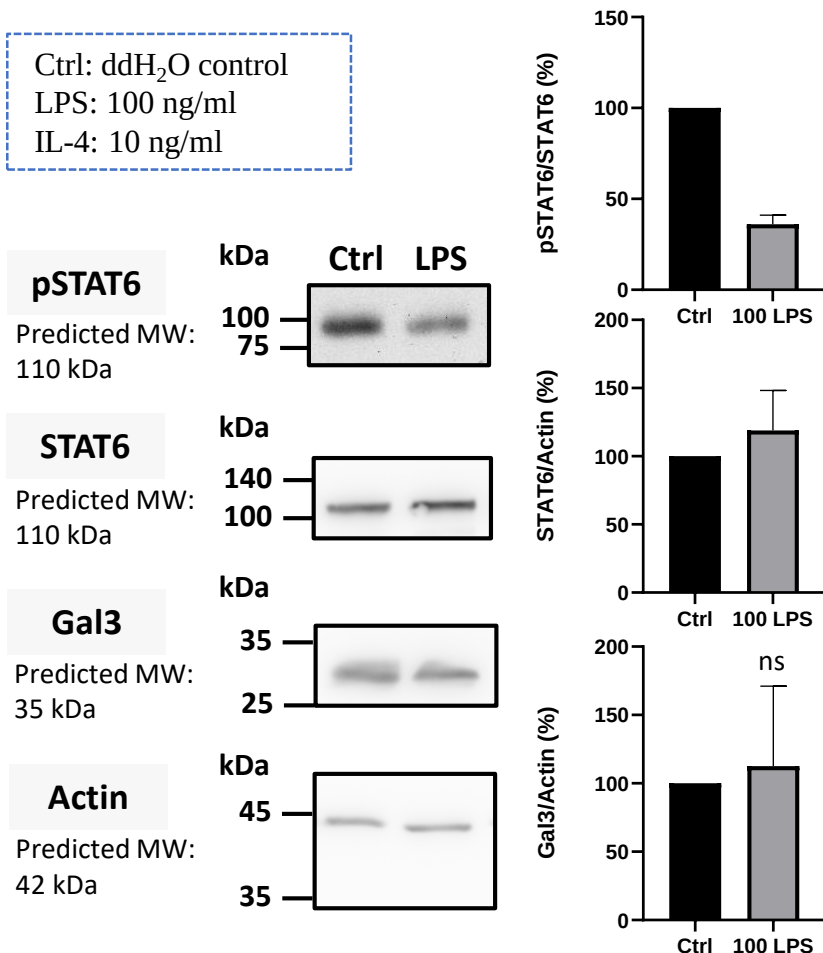
IPTG/Ctrl (%)

AT8	Tau-5
109.12	165.67
130.68	269.73
128.06	251.28
112.39	130.04

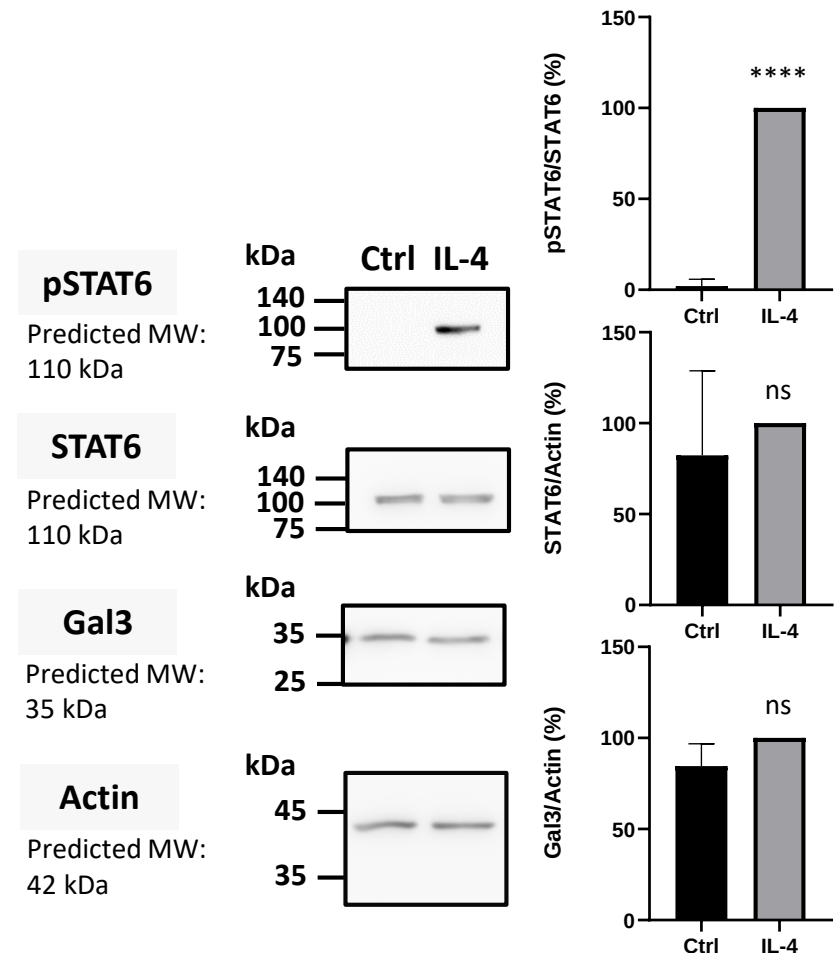
Activation of STAT6 Did Not Change Galectin-3 Expression in Naive BV2 Cells

BV2 cell seeding (1.5 x 10⁵ cells/6-cm dish) $\xrightarrow{24 \text{ hr}}$ + LPS (100 ng/ml) + IL-4 (10 ng/ml) $\xrightarrow{24 \text{ hr}}$ Collect & Lysis

Ctrl: ddH₂O control
LPS: 100 ng/ml
IL-4: 10 ng/ml

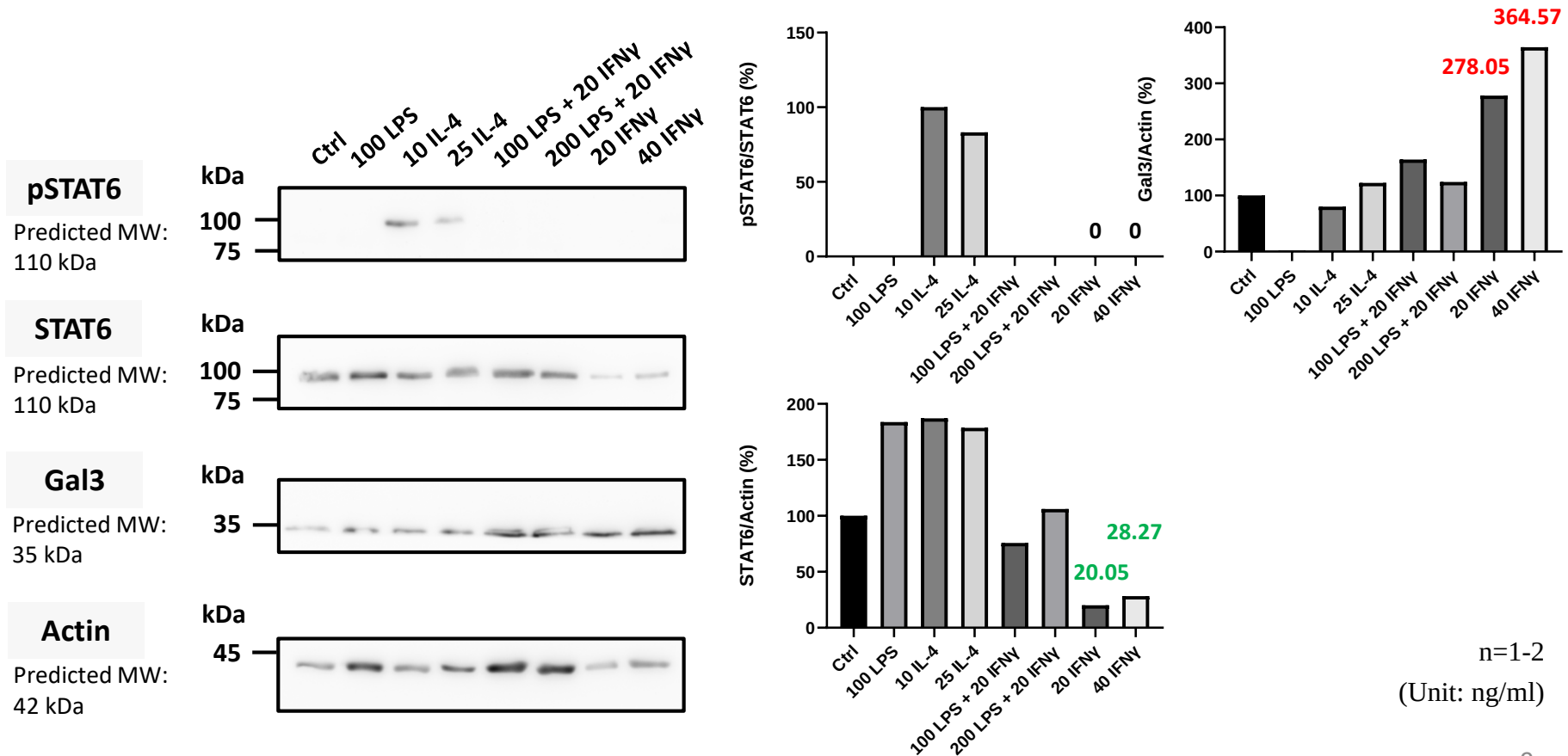


n=2-3, ns=non-significant



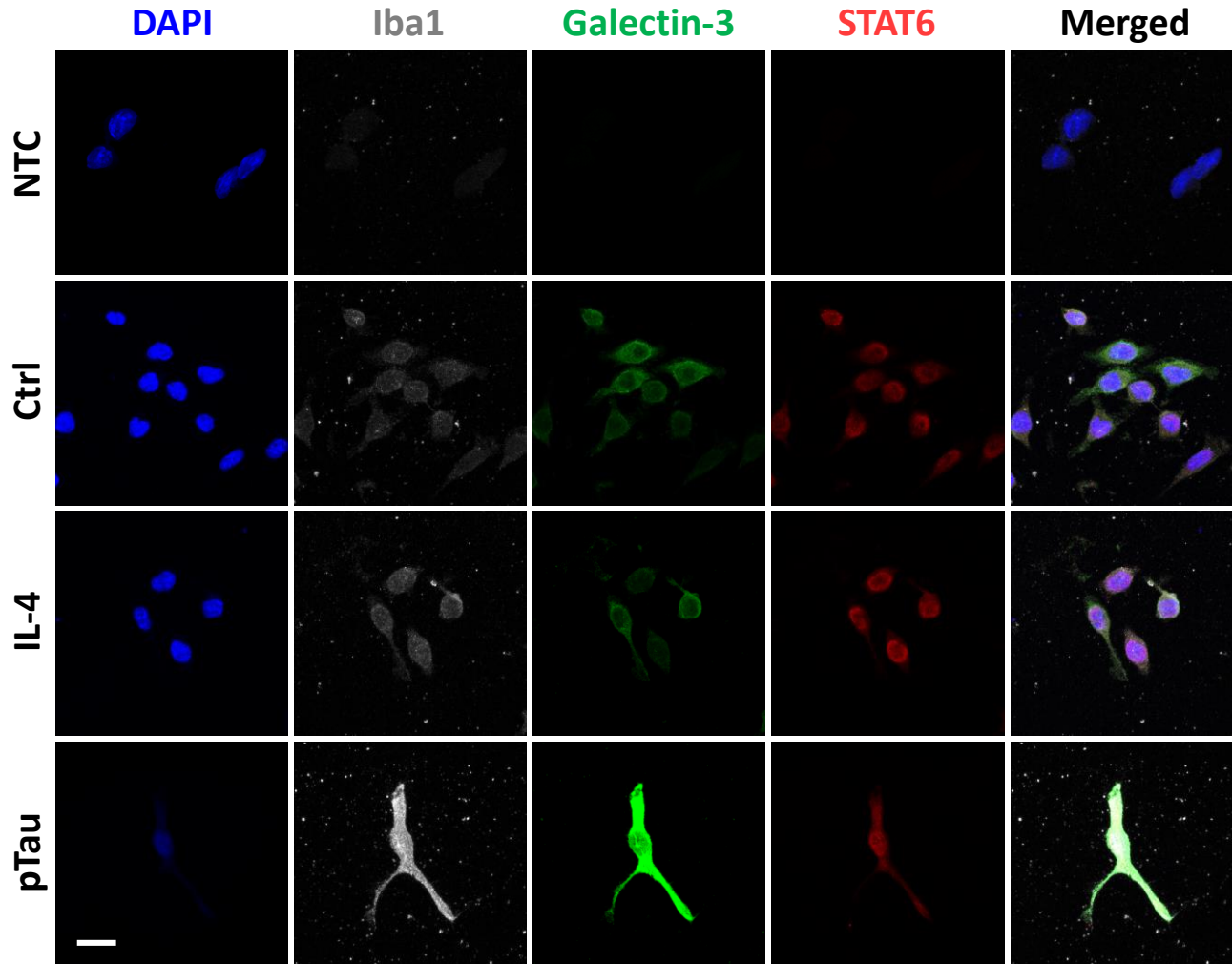
n=3, **** p<0.0001, ns=non-significant

Inhibition of STAT6 Increased Galectin-3 Expression in IFN γ -treated BV2 Cells



Galectin-3 Expression Under pTau

BV2 cell seeding
(3×10^4 cells/24-well plate) $\longrightarrow$ Treatment $\longrightarrow$ Fixation $\longrightarrow$ Mounting $\longrightarrow$ Confocal



Scale bar = 20 μ m

Conclusion

- STAT6 protein was activated in IL-4-induced BV2 cells.
- No significant change in the level of Galectin-3 in BV2 by the treatment of LPS (100 ng, 24 hr) or IL-4 (10 ng, 24 hr).
- IFN γ is likely to promote Galectin-3 expression.
- **STAT6 might inhibit Galectin-3 protein expression.**

Future Direction

- To identify whether there are other STAT6-downstream proteins that can act as Galectin-3 inhibitors.
- To treat BV2 cell with pTau and evaluate the activation of STAT6 and the expression of Galectin-3.

Acknowledgement

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